

A Short Game Theory Primer for ISS 2440

Optional reading — Week 5: The State of Nature

Fall 2026

1 What a game is

A game is any situation in which what happens to you depends on what someone else chooses as well as on what you choose. Chess is a game in this sense, but so are an arms race, a price war between two gas stations, and two roommates deciding who cleans the kitchen. Game theory is the branch of economics that studies such situations, and this week it gives us a precise way to talk about Hobbes's and Locke's state of nature.

To describe a game you need three ingredients. First, the *players*: who is choosing. Second, each player's *strategies*: the complete list of things that player can do. Third, the *payoffs*: a number for each player attached to every combination of choices, ranking the outcomes from that player's point of view — higher numbers for outcomes the player likes more. The payoffs need not be money; they stand for whatever the player cares about, years of freedom included. Game theory then asks a single question: if each player picks the strategy that serves her own payoff best, given what she expects the others to do, where does the interaction end up? In the two games below the players choose once, at the same time, without communicating. The answer to the question turns out to depend on the game — which is the point.

2 The Prisoner's Dilemma

Two suspects, Rose and Colin, are arrested for a robbery. The prosecutor has enough evidence to convict each of a minor charge worth one year in prison, but needs a confession to prove the robbery. She puts them in separate rooms and offers each the same deal: confess and testify against your partner, and I drop your charges — unless your partner also confesses, in which case you each get five years. A prisoner who stays silent while the other confesses takes the full ten years alone. Table 1 collects the four possible outcomes.

Table 1: The Prisoner's Dilemma. Entries are years in prison, Rose's first; each prisoner wants fewer years. Whatever the other does, confessing means fewer years, yet mutual confession (5, 5) is worse for both than mutual silence (1, 1).

Rose	Colin	
	Stay silent	Confess
Stay silent	1, 1	10, 0
Confess	0, 10	5, 5

Put yourself in Rose's chair and reason through Colin's two possible choices. If Colin stays silent, Rose gets one year by staying silent but walks free by confessing: confessing is better. If

Colin confesses, Rose gets ten years by staying silent but only five by confessing: confessing is better again. A strategy that beats every alternative no matter what the opponent does is called a *dominant strategy*, and confessing is dominant for Rose. The game is symmetric, so confessing is dominant for Colin too. Both confess, and both serve five years — even though both staying silent would have cost each of them only one. Notice what the argument did not use: Rose never needed to predict Colin, trust Colin, or even care about Colin. Ordinary self-interest, applied flawlessly, marched both players to an outcome both regard as worse.

Economists relabel the strategies *cooperate* (stay silent) and *defect* (confess) and find the same structure everywhere: two countries arming when both would prefer mutual disarmament, two fishing fleets racing to empty a shared stock, two students shirking on a joint project. In every case defection is individually irresistible and collectively ruinous. The dilemma is not that people are wicked; it is that the payoffs reward wickedness.

3 The Hawk–Dove game

Now change the story. Two ranchers each claim an unowned water hole worth 4 — units of value now, not prison years, so higher is better. Each can play *Hawk*, fight for the hole, or *Dove*, posture and then yield if the other fights. If both play Hawk they fight; each wins half the time, and the loser’s injuries cost 8, so a fighter’s average payoff is $(4 - 8)/2 = -2$. If Hawk meets Dove, the Hawk takes the hole (4) and the Dove walks away (0). If both play Dove, they split the water (2 each).

Table 2: The Hawk–Dove game. Entries are payoffs, the row player’s first; higher is better. Hawk is the best reply to Dove, and Dove is the best reply to Hawk, so neither strategy is dominant.

Rancher 1	Rancher 2	
	Hawk	Dove
Hawk	−2, −2	4, 0
Dove	0, 4	2, 2

Run the Prisoner’s Dilemma reasoning again and it fails. If the other rancher plays Dove, you want to play Hawk ($4 > 2$); if he plays Hawk, you want to play Dove ($0 > -2$). Neither strategy is dominant: the best choice depends on what the opponent does. To predict the outcome we need a weaker idea than dominance. A pair of strategies is a *Nash equilibrium* if each player’s strategy is a best reply to the other’s — no one can gain by changing her choice alone. The Prisoner’s Dilemma has exactly one equilibrium, mutual defection. Hawk–Dove has two in pure strategies: one rancher plays Hawk and the other plays Dove. Someone gets the water and someone backs down; the game cannot tell you who.

That indeterminacy produces the game’s two characteristic outcomes. If nothing breaks the symmetry — if neither rancher can tell who is “supposed” to yield — the stable state is a mixture of behaviors: aggression pays when it is rare (a Hawk mostly meets Doves) and fails when it is common (Hawks maim each other), so populations drift toward a blend of boldness and deference in which fights break out some of the time. But if something does break the symmetry, peace is available. Suppose everyone follows the rule “the prior possessor plays Hawk; the challenger plays Dove.” A challenger who expects the possessor to fight does best by yielding, and a possessor who expects the challenger to yield does best by standing firm, so the rule enforces itself — no referee

required. A self-enforcing rule of this kind is called a *convention*, and respect for prior possession is the oldest convention of property we know.

4 Repeated play and the shadow of the future

The one-shot Prisoner's Dilemma is bleak, but few dilemmas are played once. Suppose two merchants trade every month, and in each month's exchange honest dealing pays each 3, cheating an honest partner pays the cheat 5 and the victim 0, and mutual cheating pays 1 each — a Prisoner's Dilemma in every single month. If they trade exactly once, cheating is dominant, as before. But if no one knows which meeting is the last, a new kind of strategy becomes available: make this month's choice depend on the partner's past behavior. The simplest is *tit for tat* — deal honestly in the first month, then do whatever your partner did last month. Against such a partner, cheating buys you 2 extra this month ($5 - 3$) and costs you 2 every month afterward ($3 - 1$). If you expect the relationship to continue long enough, cheating is a bad bargain, and honesty becomes self-interested. Robert Axelrod, who ran computer tournaments in which strategies played repeated dilemmas against each other, called the mechanism “the shadow of the future”: *tit for tat*, the simplest entry, won.

Cooperation sustained this way is real but conditional. It needs a future that matters — patient players and a high chance of meeting again; it needs cheating to be visible; and it needs cheaters to be identifiable so retaliation lands on the right person. This is why stable villages, merchant networks, and repeated dealings between neighbors sustain honesty without any government, while anonymous one-shot encounters breed the behavior the one-shot model predicts. The theory even predicts its own failure: when everyone knows the last round is near, the shadow shortens and defection returns.

5 Back to the state of nature

Hobbes's state of nature is a Prisoner's Dilemma played for lives. In chapter 13 of *Leviathan*, even a man content with what he has must attack first, because others may attack him: strike-first is dominant in just the way confession is, and the equilibrium is the war of all against all — an outcome every player deplores yet no player can unilaterally escape. Hobbes's remedy is not to make men better but to change the payoffs: a sovereign who punishes defection makes covenant-keeping, not attack, the dominant strategy. That is the game-theoretic content of his line that covenants without the sword are but words — and it explains why Hobbes, unlike Axelrod's merchants, sees no exit through repeated play: where life itself is the stake, one defection can end the game before the future casts any shadow.

Locke's state of nature is Hawk–Dove. For Locke, property and its obligations exist before government, and the primer shows how that is possible: a convention of prior possession — first tiller plays Hawk, latecomer plays Dove — assigns every water hole an owner and settles contests without a fight or a sovereign. But the peace of a convention is imperfect in exactly the ways Locke lists as the “inconveniences” of his state of nature: where claims are ambiguous or possession disputed, the symmetry-breaking rule fails, both sides judge their own case, and the fight in the upper-left cell of Table 2 breaks out. Government, for Locke, is not the source of property, as it is for Hobbes; it is the umpire hired to make a mostly self-enforcing convention enforce itself all the time. Which philosopher has the better model of life without government is the question this week's readings put to you — and now you can restate it precisely: is the state of nature a Prisoner's Dilemma, or is it Hawk–Dove?